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Methods for Mobility Prediction

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Methods for Mobility Prediction

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Resumo

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Chapter 1

Introduction

Over recent years, information has grown exponentially in value. Information about mobility, location, and habits of a user is no exception. With the ever-increasing number of devices that can connect to the Internet, mobile big data has had an explosive growth. The volume of data being generated daily has increased, and the traditional methods used to process this information were no longer able to cope with such amount of data. The ability to process this data efficiently is still a necessity regardless of its growth, therefore the importance of data mining. Data mining is a key tool in data analytics and helps, in this case, to obtain information and relation between conditions that are not always that clear in a first analysis.

With knowledge of a user´s mobility, significant information about his interests, regions frequently visited and even some habits can be obtained. If you scale this to a large group of users, we can identify popular geolocation areas and that information can then be used by a wide range of services. With enough relevant information, a predictive model can be built to identify the future positions or directions of a moving node when its current position is known. However, a user may suddenly change their mobility habits, and the model must adopt fallback and aging mechanisms to enhance the prediction accuracy.

1.1 Motivation

The characterization of mobility has attracted increased interest. By properly studying and understanding mobility patterns of different types, ranging from pedestrians to bicycles or even vehicular, we can improve the quality of life and/or

services supported by mobile systems. Recent advances on vehicular traffic congestion identification allow the evaluation of congestion levels almost in real-time. The identification can be used to reroute vehicles to less congested roads, saving time and decreasing air pollution. However, several services can benefit from advanced knowledge of mobility. For example, emergency services can increase its response time if traffic lights are adequately changed to benefit emergency vehicles. Simultaneously, other traffic flows can be deviated from specific regions of a city, due to high levels of air pollution or due to the need for public emergency assistance. But mobility can also be explored in communication services. In high spatial density regions, multihop communications can be used to reach destination nodes outside the radio range. This scenario can be helpful in several scenarios, including civil catastrophe accidents.

With the advent of the Internet of Things, sensitive information can be routed to mobile nodes to increase the probability of routing data to the intended destinations. Crowdsensing is based on gathering a huge amount of data. In some scenarios, the data gathering can be performed in an opportunistic way, given the low immediacy of the data to collect. In these cases, mobile nodes (e.g. vehicles, bicycles, pedestrians, etc.) may serve as mules to transport the data to the destination nodes, or to specific nodes more suited to route the data. The scenarios above motivate the development of innovative tools to allow a deeper understanding of the mobility, including its characterization and possible ways of predicting it.

1.1.1 Objectives and Contributions

This subsection describes the objectives that were defined for this thesis as well as a few expected contributions of this work.

Objectives:

The main objective for this thesis is to design, implement and evaluate a Geolocation Prediction Architecture built on top of a mobility big data repository, which should be able to identify the next position of a moving node when its current position is known. This can be achieved by completing the following subtasks:

- Identification of popular regions by investigating different algorithms to identify regions that are often visited by the nodes. Raw geolocation data is initially processed using big data techniques, and the main objective is to mine popular regions by clustering the raw geolocation data.

- Individual trajectory modeling by designing a computational methodology to mine the trajectory of each node from the time-based sampling geolocation raw data. The individual trajectory can be viewed as a geolocation changing sequence, i.e. a list of the different popular regions visited by a node, including the arriving and departure time on each of the regions.
- Mobility prediction by investigating different techniques to predict the geolocation of each user. The probabilistic information contained in the individual trajectory will be explored in different prediction techniques that will be revisited in Chapter 2.

Expected Contributions:

- Mining information about popular geolocation areas in mobility contexts, using a clustering algorithm. This algorithm will be used to obtain the most frequently visited areas, by clustering the user´s single locations in time, based on which are visited the most.
- Provide a reliable Mobility Prediction Model, capable of determining a future position of a node, based on its previous locations. The research will be conducted to improve the balance between geolocation prediction accuracy and the algorithm complexity.

1.2 Report Structure

This report is organized as follows. Chapter 2 describes the related work in mobility enabling technologies in Information-Centric Networks (ICN), as well as the advantages that mobility characterization can bring to these networks, the most used techniques and processes to obtain information about a user's mobility, the different ways to obtain a user´s trajectory and mining popular geolocation areas and how to build a geolocation prediction model based in Markov chains, Bayesian Networks, Regressions or Convolutional Neural Networks. The different approaches are compared, and practical examples are provided for each one. Chapter 3 presents the work plan, identifying the tasks and their temporal evolution. Finally, Chapter 4 concludes the report.

Chapter 2

Related Work

2.1 Information Centric Networks and Mobility

The explosive growth of mobile data traffic leads us to believe that in a very recent future the number of mobile hosts, as well as their traffic load will largely exceed wired hosts, making host mobility an inherent issue in the current Internet Architecture [1]. According to the prediction in [2], more than 60 percent of the IP traffic generated in 2018 would have come from Wi-Fi and mobile devices. Therefore, there is a "mobility" issue in the current Internet architecture that needs to be addressed.

One of the main difficulties of overcoming the mobility issue is in the naming. The current naming system for the hosts is the Internet Protocol (IP). IP binds the name with the address, therefore, if a host's physical location changes, its name also has to change. In light of these limitations, an information-centric networking approach has been proposed to shift the current communication model to a receiver-driven content retrieval model [3, 4, 5]. In an Information Centric Network (ICN), the main focus is the content itself, not its actual location. Moreover, the naming can be now considered an information-based naming system.

2.1.1 Brief History of Information-Centric Networking

As referred in [6], ICN is an Internet architecture based in a publisher/subscriber model, in which the network's content is what matters the most to the users, not their location or from where they were obtained. ICNs' focus on supplying the

information to a user regardless of the user´s location, therefore having a great potential in solving some of the issues with the current IP networks, such as mobility, congestion, energetic efficiency, and security.

The first concept of ICN was introduced by Cheriton *et al.* in [7, 8], but since then its structure has been modified and reviewed many times. In [9] it is described a promising architecture where the communication is established in an ICN environment with the use of two different packet types: an *Interest Packet* and a *Data Packet*, that contain an identifier of the content to which they are referring to. The *Interest Packet* represents a user´s request to acquire the content he desires, and the second one is used to deliver the intended content to the user. This approach could be advantageous, improving the simplicity and leading to an easier way of achieving a mobility scenario in this type of networks.

2.1.2 ICN and its Advantages for Mobility Scenarios

Traditional IP networks represent a host-based communication model whereas an ICN architecture represents a receiver-driven approach.

The aim of an ICN is to make the content more accessible to the subscribers, by disseminating the network's contents through the hosts, rather than focusing on the communication between the end hosts itself. According to [6], the implementation of ICN requires multiple blocks, including network components such as information naming, routing, caching, security and management. The implementation of these components allows ICN based networks to have advantages in comparison with IP based networks. The advantages are related with mobility and wireless scenarios in the following perspectives:

- Host Multi-Homing: When referring to a more host-based architecture for a network, such as traditional IP networks a big limitation for mobility is that the host´s address is connected to a particular network interface, making it difficult for the same host to connect to another interfaces. On the contrary, in ICNs a different approach is used. The use of a request-reply model allows a host and its re-quests to be multiplexed over multiple interfaces, contributing to host mobility.

- Network Address Consistency: In a host-centric network, the identifier of each host (e.g. an IP address) is dependent on its physical location. If a host changes the location the communication may break down, and the host may have to be given a new identifier to resume it. These events can happen even when the information that is being transmitted between hosts is the same

(does not change over time). In an ICN perspective, given that the naming is content driven, the information can be considered content-centric. When a user requests content, he is given an identifier and routed to the best content provider.

- Connection-Less Sessions: Host-centric networks rely heavily on frequently re-establishing connection-oriented sessions to manage connections established between mobile hosts, and the sender has full control of the content transfer. In ICN a connectionless oriented approach is used to enable the transfer of content from a publisher to a subscriber, and the communication can be seen as a receiver-driven model. Consequently, when a mobile receiver moves to a different location it resends requests to obtain the rest of the data it still had not received, without the need to establish a new TCP connection or using costlier overlay solutions such as Mobile IP (MIP) [10, 11, 12].

- Replication: In host-centric networks, a concept of location is used (e.g. an URL) to transfer data. However, the process of content distribution will breakdown if a host that is a part of the delivery path fails. In ICN there is a caching ability within the network that allows for a faster and more efficient recovery, given the fact that there is a bigger "redundancy" of content scattered through the network. The processes and features that allow this type of caching to happen are described in more detail in [6] and [13].

2.1.3 Technologies and Processes that allow for Mobility in ICN

In an ICN context, it is not required that both publishers and subscribers need to be connected to the network simultaneously for the content request and delivery. The subscribed contents are momentarily transferred whenever information publications are issued, as described in [6]. Publishers and subscribers also do not need to have spatial references to each other and are not aware of how many publishers and subscribers are involved in a certain content delivery process.

This decoupling of space and time between the data transfer and the request to the transfer itself is one of the basic features of this architecture that allows for seamless mobility of mobile nodes [1, 14]. Next, we provide a brief overview of some of the technologies that allow mobility in an ICN architecture, from a subscriber and a publisher point-of-view.

2.1.3.1 Subscriber Mobility

As mentioned before, ICN has what can be called a *Native Mobility Support*, in the sense that its receiver driven model of communication, where the subscribers request the data they want by its identifier, as well as the connectionless oriented method of delivering the requested information to a user for a seamless sense of mobility. Therefore, the main issue from a subscriber viewpoint lies with the ability to perform fast retransmission when a mobile publisher moves out of reach of a subscriber, because sending more requests to retrieve the "lost" information can waste resources that could be applied in other transmissions. To resolve this issue a *Fast Request Retransmission Recovery* method is applied.

The first application of this method was introduced by Han *et al.* in [15], where a retransmission scheme that incorporated a lifetime estimation and retransmission control algorithm that can detect fluctuations in the round trip time [6], was introduced in order to solve the issue of frequent packet loss in content-centric wireless networks.

Alternatively, Amadeo *et al.* [16] propose another approach where the goal is to estimate the average round trip time to manage the retransmissions of content packets that never reached their destination in wireless content-centric networks. When an *Interest Packet* is disseminated through the network, the time information is tracked in every node that it reaches. Posteriorly, a node records a round trip time sample when it receives the data that was required. With this method, an average round trip time can be calculated each time information is received, and the round trip timeout can accompany its variations. Beyond that, a transport algorithm to address both *Interest Packet* retransmissions and traffic control in cases of undelivered packets is also proposed [17]. When a user still has not received the content it requested after retransmitting the *Interest Packet* a given number of times, he announces that content as unobtainable. Whenever a subscriber receives a non-duplicated data packet an algorithm is used that weighs the previous round trip times exponentially to update the estimate of the RTT. With this a way to adaptively adjust the RTT is obtained helping achieve better effectiveness and efficiency [6].

Other methods for improving the efficiency of the mobility of the nodes, from a subscriber perspective, in ICN environments are explained in detail in [6].

2.1.3.2 Publisher Mobility

The two main issues when referring to publisher mobility are: resolving the content provider's location whenever it changes and keeping the continuity of a session when the source is moving. Contrarily to subscriber mobility, publisher mobility represents a tougher problem to resolve, given that there is no difference between the routing locator of a content provider and the content identifier [10].

There can be considered two different approaches that address the publisher mobility issue:

- Indirection Approach: It is based on the use of home agents, whose function is to forward the *Interest Packets* to the mobile devices. To support this operation, home agents need to have information about the topology of the network. The main advantage of this approach, according to [18], is the absence of overhead in the resolution phase. However, it means that all the *Data* and *Interest packets* must have to pass through the home agent. This can give rise to some congestion problems. Within this approach there are three types of solutions that can be used:
  - *Rendezvous point*: There is a rendezvous server keeping track of the user's locations, that provides name resolution between the content identifiers and the locator after publisher mobility. A query to the rendezvous point is made whenever a subscriber does not acquire the requested content within a given period of time;
  - *Indirection point*: Uses a relay server that contains a set of naming records between a target prefix and a source prefix of a content provider. The first is a name used to serve data while the goal of the second is to temporarily receive *Interest Packets*. When an *Interest Packet* reaches the server, the server encapsulates it with the respective source prefix and routes it to the destination;
  - *Tunnel-based redirection*: Every content provider must register its own name prefix in its respective home router. This way, all the *Interest Packets* will be forwarded to the content provider's home router. When a handover occurs, and a provider switches its access point, it sends a prefix update message to its home router containing the content provider's new prefix identifier. From now on, the *Interest Packets* will be forwarded to the content router where the provider is now connected.

The previous processes are represented in Figure 2.1.

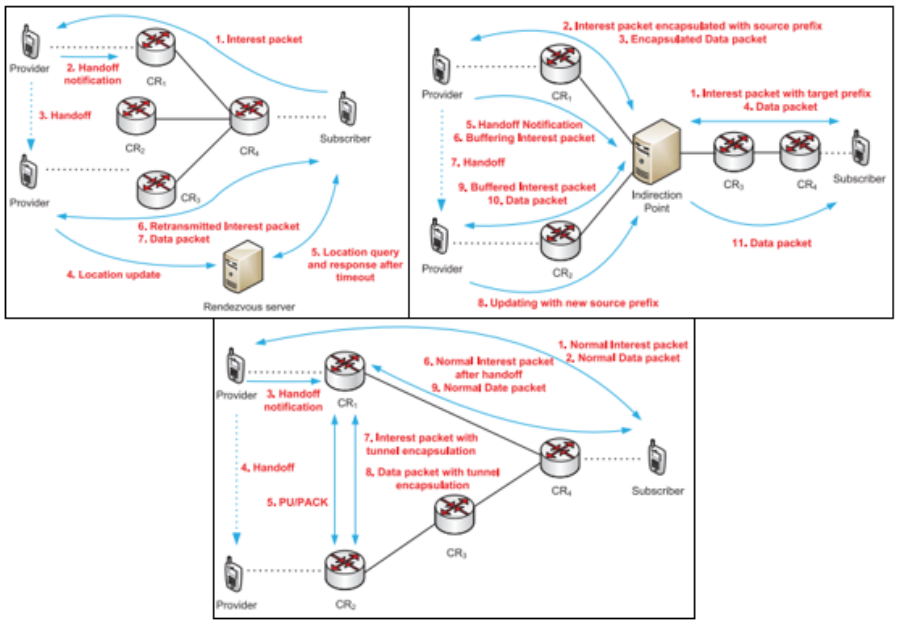

Figure 2.1: Indirection Approach to Publisher Mobility in ICN, composed by Rendezvous point (top left), Indirection point (top right) and Tunnel-based redirection (bottom) solutions (composed from schemes taken from [6]).

- Routing-Based Approach: Even though the previous approach supports mobile content providers in ICN, there are still issues that need to be addressed. These mainly concern handoff latency and overhead induced by the encapsulation of packets, affecting the QoS of the network. For this, a routing-based approach is proposed based on the following types of solutions:

  - *Flooding*: Copies of the *Interest Packets* are disseminated to the potential content routers where a moving content provider may connect. This set of routers is denominated a handoff zone, where the packets are flooded, in order to prevent a broadcast storm, as referred in [19]. The formation of these zones is dependent on the content router's location, and it can be pre-configured or learned from other network nodes;

  - *Interest forwarding*: When there is a handoff, a virtual *Interest Packet* is forwarded to the provider´s previous content router. When an intermediate router receives this packet, it updates his routing table in order to transmit the next packets to the new location. If there are any cached Interest Packets in the previous content router, a flag is set to indicate retransmission, so their loss can be avoided;

The previous processes are represented in Figure 2.2.

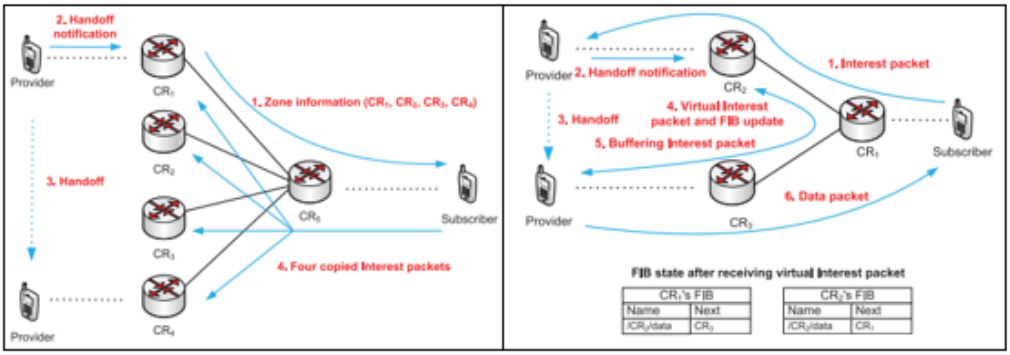

Figure 2.2: Routing-Based Approach to Publisher Mobility in ICN, composed by Flooding (left) and Interest forwarding (right) solutions (composed with schemes taken from [6]).

2.1.4 Mobility Characterization and Prediction in an ICN context

There is a new direction to be exploited in an ICN context, that is user mobility. By characterizing user mobility the whole ICN operation can be optimized by selecting better strategies in the following features: naming, routing, forwarding and caching [20] and, as mentioned above, publisher mobility still has room for improvement. Mobility can be characterized in a predictive or descriptive way.

In a descriptive approach, the focus is to describe the mobility through a stochastic model that is further simulated with the goal of creating mobility patterns similar to real trajectories. Therefore, individual or group mobility is characterized by its statistical properties and may not be adequate to describe a scenario where changes occur frequently. Furthermore, the applicability of such models is limited because the modeling approach is most of the times valid for a specific scenario, and not applicable to a different context.

In a predictive approach, the focus lies on predicting a future position in short/medium term of a mobile node considering his past mobility references and trajectories, never forgetting that the mobility habits of a user may vary, and a predictor needs to be aware of this possibility and make real-time adjustments, so the prediction accuracy is maintained. The prediction techniques often rely on linear models of mobility, where real-time tracking data is used to estimate the probability of visiting the next place. Linear prediction is only adequate if the model parameters are, in some way, adapted to the user´s mobility patterns. To estimate these parameters over time, more advanced tools are required, as well as high computing power.

In an ICN context, where there are many mobile nodes, a group mobility prediction approach may be more useful. Nodes are grouped if they show similar mobility patterns and, unlike individual prediction, unsupervised learning techniques are mainly adopted [21], having the advantage of not depending so heavily on previous user habits.

2.2 Big Data Mobility and Geolocation Prediction

Nowadays there are many services and applications that are based on user's location information. With the ever-growing number of smart mobile devices, the amount of geographic data, personal and vehicular trajectories and geolocation records generated daily is increasing. It makes sense then to refer to this data as mobility big data. As said in [22], mobility big data is defined as the geolocation data produced and collected by intelligent moving devices. However, this data has many different forms and shapes and it is not in a universal format, so we can divide mobility big data into three main types:

1. Global Positioning System (GPS)
2. Global System for Mobile Communications (GMS)
3. Wireless Fidelity (Wi-Fi) data

Adopting GPS, data is represented by the GPS coordinates (longitude and latitude) and the respective time stamp of that position. The trajectory is represented by different geographical points with different time stamps. It is the most commonly used type of data in geolocation prediction given that it is the easiest to obtain. However, in several scenarios GPS signal is not available, including but not limited to indoor environments and the user cannot turn off its receiver. When GSM is used, the data is collected through the information of the base stations and the users that are connected to them. In some cases, GSM data is difficult to obtain because the operators cannot disclose it and a user may be connected to multiple stations simultaneously. Finally, in Wi-Fi, the fingerprint method is used to obtain information, starting with the detection of a wireless signal and whether a user connects or disconnects from it. As said in [22], the response rate of a user is recorded as a fingerprint of his position, then we can deduce whether the user is in the same position by comparing the fingerprint. The difficulty of this method of data gathering is that the data can only be obtained in locations where

the quality of the signal is strong enough, making it difficult to gather mobility information in outdoor scenarios.

There are numerous benefits in mobility prediction for applications ranging from predicting group and crowd dynamics or improving the accuracy and trustworthiness of location-based recommendations. The first works in geolocation prediction go back to 2003, where Ashbrook *et al.* [23] predicted the future geolocation of the user based on its GPS history geolocation. Since then other types of research have been made in the area, being the most commonly used research methods machine learning algorithms, such as the Markov Model [24] and Bayesian Networks [25]. These methods are further described in Subsection 2.2.3.

A typical geolocation data record contains geographical coordinates and an associated timestamp, from a set of these records' popular geolocation areas and personal trajectories can be computed. The first one can be represented by a set of data points that are clustered according to the time interval and the distance difference between them. The second one can be represented by a set of popular areas ordered by time sequence.

The different steps of building a geolocation prediction model are shown in Figure 2.3 and detailed in the following subsections.

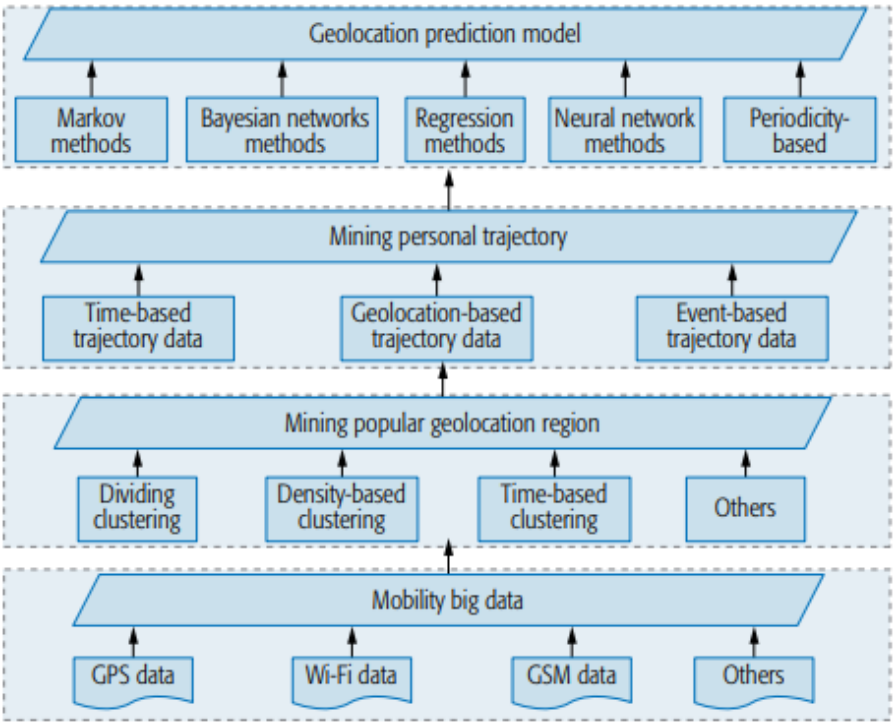

Figure 2.3: Structure and "layers" of building a geolocation prediction model, (adapted from [22]).

2.2.1 Obtaining Popular Geolocation Regions

A representation of a user's geolocation information can be either discrete or continuous. Using a representation of single points does not make it easy to obtain information about a user's popular location, therefore, they must be clustered into popular geolocation regions so that information can be extracted from them (e.g. if they represent a home location or a workspace). Considering the three main types of geolocation position data referred in Section 2.2, in what follows we briefly overview the methods for processing and obtaining popular geographical regions.

GPS Data

As mentioned before, a single geolocation position such as a pair of GPS coordinates is not very meaningful to users. These raw coordinate points must be clustered into an area, so we can assess their relevance to a user´s trajectory. Given the characteristics of this type of data, a clustering algorithm approach is a very adequate solution, as the K-means clustering algorithm. K-means clustering algorithm aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest cluster center and is described in further detail in [26]. A K-means algorithm based approach is proposed in [23], with the limitation that the number of clusters has to be predefined, conditioning the scalability of the algorithm, and making it difficult to group the outliner points. To overcome these difficulties, Kang *et al.* [27], propose a time-based clustering. In this approach, single points are clustered together if their timestamp difference is lower than a given threshold. If not, the new point is considered a new cluster. This method brings the advantage of auto-determining the number of clusters, but it can lead to an excessive number of them. To deal with this issue a density-based clustering algorithm is proposed in [28], where thresholds are used in the distance between two coordinate points and in the minimum neighborhood points of a coordinate point. A formal example is given in [22]: "*There are two thresholds: Dlimit and MinPts. Dlimit represents the distance threshold used to calculate the neighborhood of current coordinate point A. If the distance between A and another coordinate point B is less than Dlimit, B is a neighborhood point of A. MinPts is the threshold of the minimum neighborhood number of A. When the number of the neighborhood point of A is greater than MinPts, it will be marked as a new cluster, otherwise A will be marked as an invalid point*".

There are still problems that this approach does not address, such as the lack of a criterion to determine the size of a region. A way of resolving this problem

is given in [29], where the authors propose a grid-based method. The objects moving space is divided into cells that do not overlap each other. If an object passes through a cell then it is marked, and its density can be obtained keeping count of the times each cell is marked. Higher-density cells can be considered the most popular areas.

The final issue to be addressed is the lack of information due to the loss of data indoors. A way to discover indoor places in GPS data is referred in [22] by assuming that when the signal "disappears" an indoor place is considered as a circle´s radius and is treated as an indoor region if the signal reappears in this area. To further improve this process Ashbrook *et al.* [23] has considered the GPS absence time interval: when the GPS signal reappears in one of these areas within a certain period, this region is regarded as an indoor place, starting again to gather all the conditions to process GPS Data.

GSM and Wi-FiData

Most of the mobile cellular networks around the world implement location systems, where the *CellID* can be used to determine a user's location. In this solution there is no longer a problem with the division of the objects range in areas. By keeping track of a user´s trajectory we can assess how much time it spends in each cell and, in the end, compute the fingerprint of each cell. This fingerprint is composed by the cell's *CellID*, the total time users have spent in this cell and the most recent time record of the last user that has crossed this cell. With this information, clustering algorithms based on time or distance can be used in order to obtain information regarding the user's popular geolocation regions.

Regarding Wi-Fi data, these records contain the information of a user's MAC address when the connection is established with the Wi-Fi access point as well as the time the user associated to it. If a MAC address appears multiple times in the same access point, the area surrounding the access point can be considered a relevant region to the user with the corresponding MAC address.

2.2.2 Geolocation Prediction Models

According to [22], depending on the data collection methods, there are three main categories of trajectory data and methods for sampling it:

1. Time-based - Trajectory data is collected at an equal time interval.

2. Geolocation-based - Trajectory data is collected based on a change of geolocation.

3. Event-based - Trajectory data is collected by recording user behavior, such as using a mobile phone, buying a ticket, and other relevant events.

A visual description of these types of trajectory data is given in Figure 2.4 (the representation of event-based sampling of trajectory data can be given by either of the previous types).

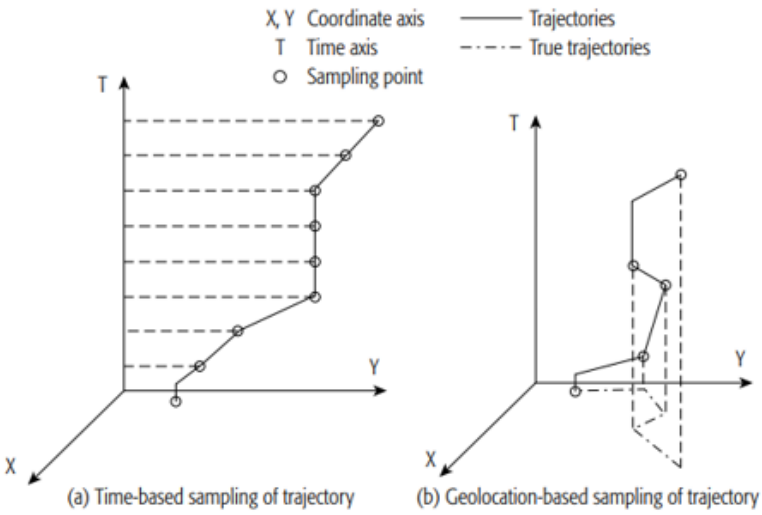

Figure 2.4: Three categories of trajectory. The "Trajectories" legend represent a three-dimensional space trajectory, and the "True trajectories" legend represents a two-dimensional space trajectory (adapted from [22]).

Time-Based Sampling of Trajectory Data

Sampling trajectory data using this method requires the use of a GPS collector that retrieves GPS information at equal time intervals. After the gathering stage this data can be represented by continuous coordinate points. In this format the trajectory is easy to obtain. However, as mentioned before, single coordinate points are meaningless to users and the popular geolocation regions need to be obtained first. After we possess this information, we must obtain the time of arrival and departure of the popular regions the user has crossed in his movement. Then we can compute a trajectory by grouping these popular regions chronologically, which can be represented in the following format

$$T = (PGR_1, PGR_2, PGR_3, \ldots, PGR_n) | PGR = (lat, long, range, t_{arrival}, t_{leave}). \quad (2.1)$$

In 2.1, PGR represents the popular regions that the user has crossed, ordered by time sequence. n represents the total popular regions of the trajectory, represented by T. Each PGR is represented by a pair of coordinates (longitude and

latitude), a range to be applied to these coordinates, and the respective arriving and departure times. It is expected that the departure time of PGR_N matches the arriving time of PGR_{N+1}. The visual representation of a user trajectory using this type of sampling is given in Figure 2.4(a). Using this approach has the benefit of producing a considerable volume of data compared to the other approaches, as well as an ample coverage of the geolocation space. However, singularity points are not representative and the amount of data can lead to redundancy.

Geolocation-Based Sampling of Trajectory Data

This type of sampling of trajectory data is more adequate to GSM and Wi-Fi based information, given that it samples the information when a change of location is made. In a GSM scenario, this can be considered a switch in *CellID*. In a Wi-Fi scenario a change of location can be considered when a user switches his connection to another Wi-Fi access point. In both processes, the time of the change can also be obtained so the conditions are gathered to represent a user's trajectory directly from the information. When compared to the previous method, this method is considerably simpler since there is only the need to collect the sampling data and then build the final trajectory according to the variation of the geolocation areas. The final trajectory can be represented by

$$T = (G_1, G_2, G_3, \ldots, G_n). \tag{2.2}$$

In 2.2, G represents a *CellID* or a Wi-Fi access point, n represents the total number of G in the trajectory, represented by T. Unlike the previous method, the trajectory is not dependent on the time feature. The trajectory is built based on the area switching sequence as the user moves through the geolocation space. The visual representation of a user trajectory using this type of sampling is given in Figure 2.4(b). The information recorded in this process is very complete but it is difficult to program a machine to build each trajectory, and doing it manually is not an option for the volume of data associated with mobility studies.

Event-Based Sampling of Trajectory Data

Using this type of sampling is similar to the time-based sampling, with the difference that the time interval between data collections is not uniform. Data is only collected when a relevant event occurs in the "life" of a user. An example is given in [22], the act of signing in to a social network or an application integrated into a smart mobile device that generates a record in the device's system logs with the semantic geolocation of a user. There is a wider field of applications for

this type of data since it can contain all the formats of position data mentioned above: Wi-Fi, GSM, and GPS. In addition, all the entries in the dataset are linked with relevant events, which means that in this type of data there is additional information that can be obtained besides only predicting geolocation. A problem of this approach compared with the other two is that the complexity of the data is higher. However, the methods used to process the data are the same as the previous types, producing a final trajectory as follows

$$T = (P_1, P_2, P_3, \ldots, P_n). \tag{2.3}$$

In 2.3, P is considered as a user position at the time a relevant event occurred, similar to PGR in 2.1 or G in 2.2. Like the previous equations, n represents the total number of P in the trajectory, represented by T. The visual representation of a trajectory using this type of sampling can be given by either (a) or (b) in Figure 2.4. Similarly to time-based sampling, there is a large volume of data being produced by this method with wide coverage of the geolocation space. However, it can be unreliable because the information being recorded is usually incomplete.

2.2.3 Geolocation Prediction Models

We have seen previously that locating and keeping track of a user´s movement is not a fundamental issue, given that there are very reliable methods to obtain it. The issue now is how can we use this information to predict a user's future position and mobility patterns. According to [22], the most used algorithms for geolocation prediction are based on Markov methods, Bayesian Networks, Regression methods, and Neural Networks. There are other implementations based on periodic methods and hidden Markov models, respectively described in [30] and [31], but they have not been used so often in this type of problems as the ones discussed before.

Markov-Based Methods

This approach is the most used in geolocation prediction problems, given that the results it produces are easy to interpret and relatively simple. The input is a user's trajectory, in the form of nodes and edges. Nodes can represent popular geolocation areas and the edges will be associated with the probability of transition between two areas that are connected. The relationship between nodes is then computed with a matrix of conditional probability with the Markov predictor of order n, based on Markov chains. The order of the model depends on the proximity of the neighbor. After all the computations a probability tree can be

put together in order to assess the probability of transition between the complete set of nodes. Although this method is frequently used in geolocation prediction situations it has his drawbacks mainly related to the model's complexity.

Increasing the order of the model will allow for more precise computing. However, it increases the algorithm's complexity significantly requiring a higher computing power to produce reliable results. Another issue of using a model based on Markov chains is that the prediction of the model is dependent on only the current state to determine the next transition, not on the previous transitions. It also ignores the effect of time, and a user's mobility is heavily dependent on the time and period of the day. Finally, when building a Markov model for geolocation prediction all the trajectories are considered independent and that does not translate to the real world, where different trajectories of a user can be correlated. For example, the trajectory home-office-gym is related to the trajectory home-office-restaurant but in a Markov model these two trajectories must be considered independent.

Implementations of this model in geolocation prediction scenarios are described in detail in [24] and [32].

Bayesian Network-Based Methods

This approach is, in short, a probability model. The first step is to transform the pure geolocation data into variables that represent coordinates and time. A part of the data is used as a training set in order to build a Bayesian network. The process of building a model is generally described in [22]:"*Confirm the interdependent related parameters according to a joint probability density equation after building a directed acyclic graph (DAG) through the interdependence relation among variables. Finally, calculate the dependency of a Bayesian model (BM) according to the transition probability equation between two neighbor times, and build a dynamic Bayesian network (DBN) by connecting local Bayesian networks in different times*"and a practical implementation of it is given in [25].

This model can effectively compute the likelihood of an event happening, but it has inherent issues with algorithm complexity and difficult computing. It has scalability issues given that it is a more static model and relies heavily on the quality of the data, and the accuracy decreases when the records have missing data.

Regression-Based Methods

When considering this method, it is required for the input to be the user's geolocation position and its respective timestamp represented by a tuple $I =$

[$latitude, longitude, timestamp$]. The model is then built on top of the regression function that emerges from the maximum likelihood of events based on the characteristics of the data. After this, the model receives a time of a geolocation prediction as input and the output is a prediction of the user's geolocation region at that future time. All types of regression functions can be used, from linear and nonlinear regression to multivariate regression.

An issue related to this approach is the difficulty to adapt it to real-world scenarios and data. The prediction accuracy can only be maintained in reasonable values when the predictions being made by the model are short-term predictions and when the mobility patterns are ideal, which requires consistent movement direction and rate. A practical implementation of this model in a big data mobility scenario is given in [33].

Neural Network-Based Methods

When using this type of models, the trajectory of a user is split into two segments. And for training the network the first segment is given to the predictor as an input and the output will be the prediction of the remainder of the trajectory. The layers and nodes of the network are then adjusted according to the accuracy of the predictions in the training set. To maintain the accuracy at a reasonable level the dimension and variety of the dataset must be considerable so the algorithm can be adjusted considering all the possible scenarios. An inherent limitation of this approach is that a model built using a neural network only has the ability to predict the future mobility of a single user, making it hard to adapt to real world situations. An implementation is given in [34]. However, not all the implementations of models based on neural networks follow this structure. In [35] a geolocation prediction model based on a Convolutional Neural Network is proposed. The network itself is scalable to new input features or merge existing ones according to the dataset, in order to improve the model's accuracy and reduce overfitting. The network's structure is divided into two smaller networks, the first one makes a crude prediction in a wider area and based on that prediction the second sub-model executes a fine prediction to a more specific area.

Chapter 3

Work Plan

3.1 Task Description

The work plan of this thesis includes the following tasks:

1. **Identification of Mobility Scenarios**
   This task aims to identify different repositories of mobility traces that will support the methodologies developed in the thesis. The repositories will be analyzed to identify the types of mobility they contain, the way how the mobility is described (data), and a preliminary assessment of the feasibility of using them in the context of this work.

2. **Characterization of Mobility Patterns**
   The objectives of this task are the offline characterization of the mobility data. Nodes' location can be analyzed first in the spatial domain. Temporal information is analyzed further to obtain a spatial-temporal distribution of the nodes over the region of interest. Moving directions and transition probabilities between predefined locations are of interest. Different tools are expected to be adopted, including data mining tools as well as traditional statistical methods.

3. **Mobility Estimation**
   This task aims to predict the level of mobility. The goal is to test appropriate predictive algorithms capable of predicting future locations of the nodes. Different approaches are planned to be implemented. These may include

Markov Chains, Hidden Markov Chains, Bayesian Networks and Neural Networks.

4. **Technical Reporting and Thesis Writing**
   This task includes all activities related with technical reporting (description of the methods adopted for the characterization and estimation of mobility and performance evaluation) and writing of the thesis.

3.2 Schedule

The schedule that was defined for this thesis is described in Figure 3.1.

ID	Task Name	Start	Finish	Duration
1	Identification of mobility scenarios	18/02/2019	08/03/2019	3w
2	Characterization of Mobility Patterns	18/02/2019	30/04/2019	10,4w
3	Mobility Estimation	01/04/2019	31/07/2019	17,6w
4	Technical Reporting and Thesis Writing	01/03/2019	30/08/2019	26,2w

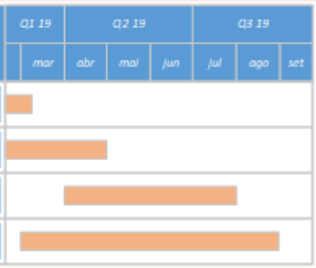

Figure 3.1: Gantt chart regarding the proposed schedule

Chapter 4

Conclusions

This aims to implement a Geolocation Prediction Architecture built on top of a big data repository, also aiming to compare the results obtained in mobility prediction scenarios using different methods such as Markov Chains, Hidden Markov Chains, Bayesian Networks and Neural Networks. The goal is to investigate in a new direction, where we look to exploit user mobility to improve the operation of an Information-Centric Network by improving and selecting better strategies in naming, routing, forwarding and caching. Publisher mobility raises multiple issues that can be addressed if a proper characterization of mobility is made, based on a predictive approach.

When we look deeper into the mobility aspect of this work, most of the presented methods are reliable and trustworthy. However, there are still some issues related with obtaining mobility information and mobility prediction. One of them being the missing mobility data, mostly caused by the loss of GPS signal in indoor environments. This will affect the geolocation prediction directly because using discrete regions when intermediate data is missing instead of continuous geolocation points will cause a decrease in accuracy of the prediction models.

Being aware of these issues, the first step will be to identify what are the mobility scenarios present in the big data repositories and determine if they have enough quality to support the methodologies that we will try to develop in this thesis. Following that will be the characterization of mobility patterns aiming to produce a spatial-temporal distribution of the mobile nodes, ultimately leading to test predictive algorithms over this information to predict the future mobility of these nodes.

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